

Aryan Kakkar

647-624-7486 | kakk2492@mylaurier.ca | linkedin.com/in/aryan-kakkar | github.com/aryankakk | aryankakkar.vercel.app/

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, JavaScript, HTML, CSS, TypeScript, VBA, PowerShell
Data & Modeling: NumPy, pandas, SciPy, scikit-learn, PyTorch, time-series analysis, Monte Carlo simulation
Visualization & APIs: Matplotlib, Streamlit, Chart.js, FastAPI, Flask
Developer Tools: Git/GitHub Actions, JupyterLab, Visual Studio Code (VSCode), JetBrains, Jest, Figma
Certificates: Java Object-Oriented Programming, Learning Java 11 (LinkedIn Learning)

EDUCATION

Wilfrid Laurier University Sep. 2023 – Aug. 2027
Honours Bachelor of Science in Computer Science (Co-op), Minor in Economics Waterloo, ON

- Concentration in Big Data Systems
- Relevant Coursework: Probability & Statistics (R), Linear Algebra, Algorithms, Data Structures, Machine Learning, Databases, Data Visualization, Canadian Banking & Financial Systems

EXPERIENCE

Machine Learning Engineer Intern Sep. 2025 – Dec. 2025
Multiverse Computing Toronto, ON

- Designed probabilistic time-series forecasting models using XGBoost, quantile regression, and transformer-based approaches to evaluate multi-season performance under uncertainty.
- Built Monte Carlo simulation pipelines with Gaussian uncertainty modeling to generate outcome distributions, confidence intervals, and scenario risk across forecast horizons.
- Improved model efficiency and deployment readiness by applying model compression techniques, including layer pruning and teacher-student (distillation) training, while preserving predictive accuracy.
- Optimized loss functions and evaluated model stability across simulated paths to balance bias-variance trade-offs and improve robustness.
- Enhanced model interpretability through clustering, feature importance analysis, and quantile-based visualizations to explain key drivers and outcome paths to stakeholders.
- Trained and evaluated forecasting models using Multiverse Computing's Singularity platform, integrating probabilistic outputs into downstream analytics and reporting workflows.

Summer Camp Counselor & BACP Supervisor Jul. – Aug. 2023 & 2024
City of Richmond Hill – Rouge Woods & Richmond Green Community Centers Richmond Hill, ON

- Coordinated scheduling and logistics for programs serving up to 100 participants, optimizing daily operations.
- Adjusted program planning using participant feedback and attendance trends to improve flow and engagement.

Customer Service Representative Aug. 2021 – Aug. 2022
Farm Boy Inc. Richmond Hill, ON

- Delivered customer-facing recommendations and supported inventory and store operations through collaboration.

PROJECTS

Personal Portfolio | *React, JavaScript, HTML/CSS, GitHub, Git*

- Designed and built a responsive personal portfolio to showcase projects, experience, and technical skills.
- Improved usability and performance through iterative testing, responsive design, and accessibility-focused UI updates.

Budget Planner | *JavaScript, React.js, Chart.js, CSS, GitHub, Git*

- Built an interactive financial dashboard to visualize spending patterns and savings scenarios.
- Calculated monthly totals and category breakdowns to generate charts for budgeting insights.

Clicker Game | *JavaScript, HTML/CSS, GitHub, Git*

- Developed a browser-based game with real-time interaction, dynamic difficulty scaling, and responsive design.
- Iterated features based on testing feedback to improve engagement and performance.

EXTRACURRICULARS

Director of Logistics Sep. 2025 - Present
Laurier Cricket Club Waterloo, ON

- Streamlined club operations by improving workflows and coordinating with all departments.
- Supported VP Operations in ensuring smooth execution of events, matchdays, and internal processes.

Project Developer Sep. 2024 - Sep. 2025
Laurier Cybersecurity Club Waterloo, ON

- Built full-stack analytics features for a MERN platform, integrating REST APIs to display real-time data visualizations for event tracking and user metrics.
- Integrated backend APIs with front-end dashboards for real-time metrics.
- Contributed to Agile sprints, debugging, and peer code reviews in GitHub.

Community Representative - First Year Leadership Program Sep. 2023 – May. 2024
House Council – Wilfrid Laurier University Waterloo, ON

- Coordinated events and served as liaison between students and residence leadership.
- Promoted campus-wide digital campaigns and initiatives using data-driven outreach.